Computer Vision
Exam A

April 2026

Name, Surname, Student ID [please compile here]:

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The exam has to be carried out in 1 hour.

- **Exercise 1.** [8 points]
  Describe the Sobel filter for edge detection. How does it differ from the Canny edge detector?

- **Exercise 2.** [8 points]
  Explain feature matching and how the RANSAC (Random Sample Consensus) algorithm can be used in conjunction with feature matching to improve the accuracy of homography estimation between two images.

- **Exercise 3.** [8 points]
  a. Explain the concept of fundamental matrix. How is it computed, and what information does it convey about the relationship between two images?
  b. What is the geometric meaning of the epipoles in the two images? How are they related to the fundamental matrix (algebraically)?

- **Exercise 4.** [8 points]
  Propose a method to detect keyframes in a video sequence using traditional (non-deep learning) techniques. Explain the features you would extract, how you would measure frame importance or uniqueness, and the criteria for selecting keyframes. Justify each step in your approach. A keyframe in this case could be: a frame that is visually different from surrounding frames (e.g., scene changes, new objects) or a frame where meaningful action begins (e.g., explosion, expression change, speaker switch).